

Kunal Agrawal

Portfolio
Github

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LinkedIn

PROFESSIONAL SUMMARY

AI/ML Engineer skilled in Python, machine learning, deep learning, and LLM-powered applications, with hands-on experience in RAG and AI agents. Experienced with LangChain, LangGraph, HuggingFace, vector databases, model APIs, and AI/ML deployment using FastAPI and Docker. Strong in building end-to-end intelligent systems, from data processing and model development to production-ready AI applications.

EDUCATION

IMS Engineering College

Ghaziabad, India

B.Tech - Computer Science and Engineering — GPA: 7.81

October 2023 – June 2027

Courses: Data Structures and Algorithms, Artificial Intelligence, OOPs, Computer Networks, DBMS, Operating Systems

PROJECTS

Incident Intelligence Agent — AI-Powered Root Cause Analysis

July 2026

LangGraph, FastMCP, PGvector, LiteLLM, Langfuse, Guardrails AI

GitHub

- Architected a production-grade Agentic RAG system to autonomously investigate production incidents, targeting root cause analysis time reduction from **~45 minutes to under 30 seconds**.
- Engineered a hybrid retrieval pipeline using **pgvector HNSW, PostgreSQL Full-Text Search, Reciprocal Rank Fusion (RRF)**, and recency-aware ranking; validated on **150 incidents across 15 microservices**, achieving **0.62–0.78** top-1 cosine similarity.
- Developed a self-correcting multi-agent workflow with **3 FastMCP tools**, citation verification, Guardrails AI, LiteLLM routing, and Langfuse/RAGAS integration for reliable, observable, evidence-backed reports.

DocMind — Offline Multimodal RAG System

June 2026

Python, FastAPI, Node.js, Qdrant, Whisper, CLIP, Ollama

GitHub

- Built a hybrid retrieval engine fusing Qdrant dense search with BM25 sparse search via Reciprocal Rank Fusion (RRF) — combining 2 embedding models and 2 retrieval strategies for higher recall over single-modal baselines.
- Ingested 3 modalities (PDF, images, audio) in a unified pipeline — Tesseract OCR for image-to-text, Whisper for timestamped audio chunking — enabling cross-modal semantic search across all document types.
- Served offline LLM inference via Ollama (Qwen2.5 / Phi-3) with 0 cloud API calls, containerized via Docker Compose (Qdrant + Ollama) for single-command reproducible deployment on air-gapped systems.

Skribbl.io Clone — Multiplayer Drawing & Guessing Game

May 2026

MERN, TypeScript, Socket.IO

GitHub — Live Demo

- Built a full-stack multiplayer drawing and guessing game with the MERN stack and TypeScript, implementing authentication, room management, and persistent game sessions.
- Engineered real-time canvas synchronization, chat, word guessing, timers, and game-state updates using Socket.IO, enabling seamless concurrent gameplay across multiple rooms.
- Designed an event-driven backend using OOP principles with disconnect recovery, host migration, and vote-kick mechanisms for resilient multiplayer session management.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java
- **ML & DL:** NumPy, Feature Engineering, Model Evaluation, Classification, Regression, TensorFlow, PyTorch, Neural Networks, Transfer Learning
- **AI & GenAI:** LLMs, RAG, LangChain, LangGraph, RAGAS, HuggingFace, Prompt Engineering, Fine-Tuning, AI Agents
- **AI APIs & Models:** OpenAI API, Groq API, Ollama, Qwen, LLaMA, Mistral, Whisper, CLIP
- **Retrieval & Agentic AI:** MCP, Qdrant, ChromaDB, Pinecone, Embeddings, Semantic Search, RRF Hybrid Search
- **Databases & MLOps:** PostgreSQL, MongoDB, MySQL, Docker, Git, GitHub, FastAPI, REST APIs

CODING PROFILES

- **LeetCode** – 250+ Problems — max: 1523
- **CodeChef** – 200+ Problems — max: 1346 — 2 Star
- **Codeforces** – 120+ Problems — max: 817

ACHIEVEMENTS

- Smart India Hackathon 2025 – **National Finalist** (Top 300 out of 50,000+ participating teams).
- Solved **400+ Data Structures and Algorithms problems** covering **Arrays, Stacks, Linked Lists, Queues, Trees, and Recursion** across competitive programming platforms.